Opgaver uge 3 / Kapitel 4

Klassisk Fysik

Opgaver, kapitel 4

4.50

46. A 2.50-kg object is moving along the x-axis at 1.60 m/s. As it passes the origin, two forces $\vec{F}_1$ and $\vec{F}_2$ are applied, both in the y-direction (plus or minus). The forces are applied for 3.00 s, after which the object is at $x = 4.80$ m, $y = 10.8$ m. If $\vec{F}_1 = 15.0$ N, what's $\vec{F}_2$?

4.51

51. Blocks of 1.0, 2.0, and 3.0 kg are lined up on a frictionless table, as shown in Fig. 4.21, with a 12-N force applied to the leftmost block. What's the magnitude of the force that the rightmost block exerts on the middle one?

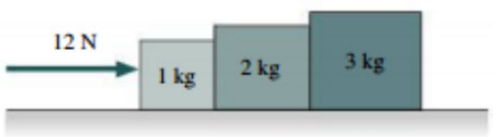

FIGURE 4.21 Problem 51

4.58

54. A 2.0-kg mass and a 3.0-kg mass are on a horizontal frictionless surface, connected by a massless spring with spring constant $k = 180$ N/m. A 15-N force is applied to the larger mass, as shown in Fig. 4.24. How much does the spring stretch from its equilibrium length?

4.62

60. Your engineering firm is asked to specify the maximum load for the elevators in a new building. Each elevator has mass 480 kg when empty and maximum acceleration 2.29 m/s^2. The elevator cables can withstand a maximum tension of 21.0 kN before breaking. For safety, you need to ensure that the tension never exceeds two-thirds of that value. What do you specify for the maximum load? How many 75-kg people is that?

4.73

73. Two masses are joined by a massless string. A 30-N force applied vertically to the upper mass gives the system a constant upward acceleration of 3.2 m/s^2. If the string tension is 18 N, what are the two masses?

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